

Core and valence photoemission spectra of atoms and molecules from a multichannel Dyson equation

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We recently presented multichannel Dyson equations for the *ab initio* simulation of various spectroscopies. In particular, we introduced a multichannel Dyson equation for the description of photoemission spectra. In this work, we apply our approach to the simulation of photoemission spectra of atoms and molecules. We introduce a numerically efficient approach to calculate their spectral functions. We compare the spectra obtained within the multichannel Dyson equation to those obtained with full configuration interaction and the *GW* method. We are thus able to show that the satellite features due to shake-up processes are significantly better described by the multichannel Dyson equation than by *GW*. Finally, we also discuss the slow convergence of the satellite energies with the size of the basis set and we propose a simple extrapolation method to reach the complete basis-set limit.

I. INTRODUCTION

Ultraviolet and X-ray photoemission spectroscopy (UPS and XPS) are powerful experimental techniques used to probe the electronic structure of atoms and molecules by measuring the kinetic energy of electrons ejected upon irradiation with high-energy photons [1]. While UPS primarily accesses valence electrons and is widely used to study chemical bonding and molecular orbitals, XPS probes core-level electrons, providing element-specific information and insight into oxidation states and local chemical environments. These methods are invaluable in chemistry because they give direct experimental access to ionization energies and electronic structure, helping to test and refine theoretical models. A key feature of photoemission spectra is the presence of satellite structures, which are additional peaks beyond the main ionization lines, that arise from electron correlation effects. In particular, shake-up processes, where ionization is accompanied by the excitation of another electron, produce distinct satellites that encode detailed information about many-body interactions and electronic relaxation, making them essential for a deeper and more accurate interpretation of complex systems.

From a theoretical perspective, UPS and XPS spectra are most often simulated using Green's function approaches since the poles of the 1-body Green's function correspond to the electron removal energies which includes both quasi-particle and satellite energies. The most common Green's function approach to simulate UPS and XPS spectra is the *GW* method [2–7]. While *GW* can accurately predict quasi-particle energies, it fails to correctly describe satellite energies. In particular, *GW* greatly overestimates the binding energies of the satellites [8–12]. We note that the *GW* method also suffers from other shortcomings [13–18]. Combining the *GW* method with the cumulant expansion approximation generally improves the description of the satellites [9, 10, 19–

21]. However, when applied to atoms and molecules it introduces unphysical excitations [22]. Wave-function methods such as configuration interaction and coupled-cluster theory can also produce accurate quasi-particle and satellite energies [23–31]. Finally, wave-function theories can be used to obtain the one-body Green's function through its definition in terms of many-body wave functions, thus allowing for the calculation photoemission spectra [32–38]. However, all these wave-function approaches are numerically more expensive than *GW* and are, therefore, limited to small molecules.

In recent years, we have developed the multichannel Dyson equation (MCDE) in which two or more many-body independent-particle Green's functions are coupled through a multichannel self-energy [11, 39–42]. In particular, for photoemission spectroscopy, we introduced a MCDE that couples the Hartree-Fock one-body-Green's function (1-GF) to the 2-hole-1-electron (*2h1e*) and 2-electron-1-hole (*2e1h*) channels of the Hartree-Fock three-body Green's function (3-GF)[43]. While the 1-GF contains information about quasi-particles the *2h1e* and *2e1h* channels of the Hartree-Fock 3-GF contain information about satellites, thereby putting quasiparticles and satellites on equal footing. We have recently successfully applied our approach to the *ab initio* simulation of the direct and inverse photoemission spectrum of bulk silicon [44].

In this work we extend the MCDE to the simulation of the UPS and XPS spectra of atoms and molecules and we will assess its quality by comparing to spectra obtained using full configuration interaction (FCI) and *GW*. We will also describe a numerically efficient approach to obtain the spectral function from the MCDE. Finally, we will demonstrate the extremely slow convergence with the size of the basis set of the satellites in the spectral functions. To solve this problem, we propose a simple extrapolation method to reach the complete basis set limit for both quasiparticle and satellite energies.

The article is organized as follows. In section II we give a reminder of the MCDE equations one has to solve in practice. Then, in section III, we discuss the details of our implementation and in section IV we give some computational details. In section V we report the spectral functions for atoms and molecules we obtained with the multichannel Dyson equation

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and compare them to FCI and *GW* spectra. We also show how to obtain MCDE spectral functions in the complet basis set limit. Finally, in section VI we draw conclusions from our work and give an outlook on future developments.

II. THEORY

To describe photoemission spectra, we will use the (3,1)-MCDE which describes the removal or addition of an electron by coupling the Hartree-Fock 1-GF to the $2h1e$ and $2e1h$ channels of the Hartree-Fock 3-GF [11, 39]. It is convenient to rewrite the (3,1)-MCDE as an eigenvalue problem of an effective Hamiltonian. The latter is given by

$$H_3^{\text{eff}} = \begin{pmatrix} H^{1p} & H^{1p/3p} \\ H^{3p/1p} & H^{3p} \end{pmatrix}, \quad (1)$$

in which

$$H_{i;m}^{1p} = \varepsilon_i \delta_{im}, \quad (2)$$

$$H_{i;mok}^{1p/3p} = (f_m - f_k)(f_o - f_k) \Sigma_{i;mok}^{1p/3p}, \quad (3)$$

$$H_{ijl;m}^{3p/1p} = (f_i - f_l)(f_j - f_l) \Sigma_{ijl;m}^{3p/1p}, \quad (4)$$

$$H_{i>jl;m>ok}^{3p} = (\varepsilon_i + \varepsilon_j - \varepsilon_l) \delta_{im} \delta_{jo} \delta_{lk} + (f_i - f_l)(f_j - f_l) \Sigma_{ijl;mok}^{3p}. \quad (5)$$

where f_i and ε_i are the occupation number and energy, respectively, of the Hartree-Fock spin-orbital φ_i . At $T = 0$ K, which is the case treated here, the occupation numbers are 1 for occupied orbitals and 0 for unoccupied orbitals. The conditions on the spin-orbitals $i > j$ and $m > o$ in Eq. (5) prevent the double counting of contributions. The various contributions to the the multichannel self-energy are given by

$$\Sigma_{ijl;mok}^{3p} = [(1-f_i)(1-f_j)f_l - f_i f_j (1-f_l)] [\delta_{lk} \bar{v}_{ijom} + \delta_{mj} \bar{v}_{iklo} + \delta_{io} \bar{v}_{jklm} - \delta_{oj} \bar{v}_{iklm} - \delta_{im} \bar{v}_{jkl o}], \quad (6)$$

$$\Sigma_{i;mok}^{1p/3p} = \bar{v}_{ikom}, \quad (7)$$

$$\Sigma_{ijl;m}^{3p/1p} = \bar{v}_{ijlm}, \quad (8)$$

where $\bar{v}_{ijlm} = v_{ijlm} - v_{ijml}$ with the matrix element of the Coulomb interaction $v(\vec{r}_1, \vec{r}_2)$ given by

$$v_{ijlm} = \iint dx_1 dx_2 \varphi_i^*(x_1) \varphi_j^*(x_2) v(\vec{r}_1, \vec{r}_2) \varphi_l(x_2) \varphi_m(x_1). \quad (9)$$

The occupation numbers in Eq. (6) ensure that the hole-hole-electron and electron-electron-hole channels do not mix in Σ^{3p} . While Σ^{3p} dresses the three independent particles (2 holes and 1 electron or 1 hole and 2 electrons) in the 3-particle channel, $\Sigma^{1p/3p}$ and $\Sigma^{3p/1p}$ couple the 3-particle channel to the 1-particle channel, thus dressing the independent particles in the 1-particle channel. We note that, within this approximation to the multichannel self-energy, the effective Hamiltonian in Eq. (1) is equivalent to that of the 2-particle-hole-Tamm-Dancoff approximation [46].

From the eigenvalues E_λ and eigenvectors A_λ of the effective Hamiltonian H_3^{eff} we can construct the one-body Green's function according to

$$G_{i;m}(\omega) = \sum_\lambda \frac{A_\lambda^i A_\lambda^{*m}}{\omega - E_\lambda + i\eta \text{sgn}(E_\lambda - \mu)} \quad (10)$$

in which μ is the chemical potential and η is a positive infinitesimal. Therefore, the spectral function, which corresponds to the imaginary part of the one-body Green's function, can be obtained from

$$A(\omega) = \sum_i \sum_\lambda A_\lambda^i A_\lambda^{*i} \delta(\omega - E_\lambda) \quad (11)$$

where i is restricted to the 1-particle channel.

III. IMPLEMENTATION

The effective Hamiltonian in Eq. 1 is expressed in terms of the spin-orbitals. Therefore, for closed-shell systems, the three-particle channel contains contributions from both spin doublets and spin quartets. However, only the doublet contributions couple to the one-particle channel since the latter only contains spin doublets. As a consequence, we transformed the Hamiltonian such that doublet and quartet states are separated. The details of the spin adaptation of the Hamiltonian are given in Appendix A. The size N^{SA} of the effective Hamiltonian for doublet states only is given by

$$N^{SA} = 2 \left[n_V \frac{n_O(n_O + 1)}{2} + n_O \frac{n_V(n_V + 1)}{2} \right] + n_{Bas}, \quad (12)$$

where n_O and n_V are the number of occupied and virtual spatial orbitals, respectively, and $n_{Bas} = n_O + n_V$. In Eq. (12), the expression in brackets on the right-hand side is the total number of unique 3-particle states. It is multiplied by 2 because 2 distinct spin doublets exist for each three-particle state. Finally, n_{Bas} is the total number of 1-particle states.

It is numerically too expensive to diagonalise the effective Hamiltonian. Therefore we use the Haydock-Lanczos method to calculate the spectral function [47]. This method follows the standard Lanczos algorithm in which the effective Hamiltonian is transformed into a real symmetric tridiagonal matrix T with diagonal elements a_i and off-diagonal elements b_i . However, instead of diagonalising the tridiagonal matrix, we obtain the spectral function directly from a continued fraction according to

$$A(z) = \frac{1}{\pi} \text{Im} \left[z - a_0 - \frac{b_1^2}{z - a_1 - \frac{b_2^2}{z - a_2 - \dots}} \right]^{-1}, \quad (13)$$

with $z = \omega + i\eta$. It is important to note that, in order to select the one-particle channel, the initial Lanczos vector should have non-zero elements only in the one-particle space.

Since the spectral function is obtained using the Lanczos-Haydock method, the bottleneck of a (3,1)-MCDE calculation

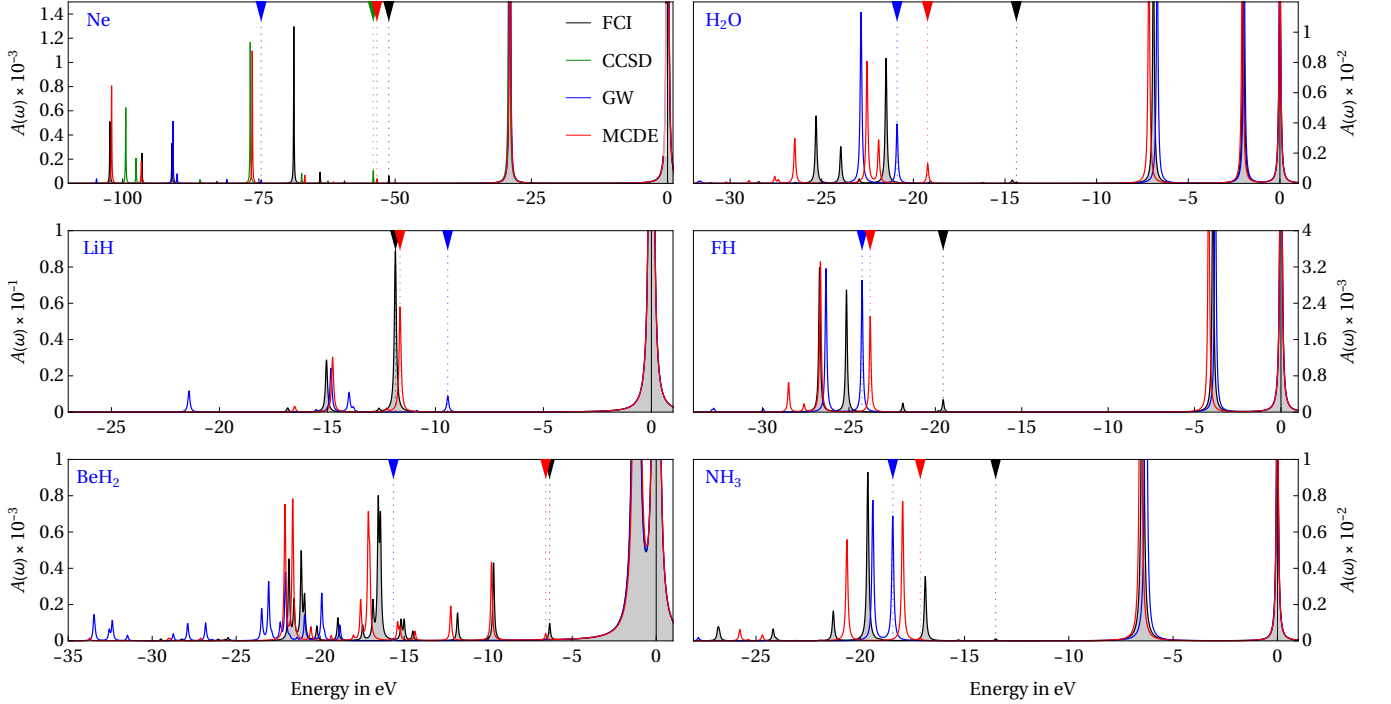


FIG. 1. The valence spectral functions of Neon and various small molecules calculated within FCI (black), GW (blue), and MCDE (red). For Neon, we also report the GFCCSD (green) spectral function taken from Ref. [45]. The arrows indicate the satellites with the lowest binding energy for each method.

is the construction of Σ^3p in the effective Hamiltonian. The calculation of the hole-hole-electron channel of Σ^3p scales as $n_O^4 n_V + n_V^3 n_O^2$ while the calculation of its electron-electron-hole channel scales as $n_V^4 n_O + n_O^3 n_V^2$. Therefore, the overall scaling of a (3,1)-MCDE calculation is N^5 with N the number of electrons in the atom or molecule.

IV. COMPUTATIONAL DETAILS

The geometries of the molecules were taken from Ref. [48]. All Hartree-Fock calculations were performed with PySCF [49]. The Hartree-Fock orbital energies as well as the two-electron integrals in Hartree-Fock orbitals were then read into our MCDE code to obtain the spectral functions. The GW spectral functions were obtained with MolGW [50]

The FCI calculations for the removal part of the spectral function were performed with PySCF. First, we calculated the FCI ground-state wave function $|\Psi_0\rangle$ and corresponding energy E_0^N of the N -electron system. Then, we calculate all ground and excited-state wave functions $|\Psi_i^{N-1}\rangle$ and corresponding energies E_i^{N-1} of the system with $N-1$ electrons. The removal energies and corresponding transition amplitudes are then calculated as $E_i^{N-1} - E_0^N$ and $\sum_j |\langle \Psi_i^{N-1} | a_j | \Psi_0^N \rangle|^2$, respectively.

In order to extrapolate the satellite energies to the complete basis set limit, one has to keep track of the excitation each

satellite corresponds to in the various basis sets. This information is stored in the eigenvectors of the effective Hamiltonian in Eq. (1) which contain coefficients corresponding to the one- and three-particle excitations. We obtain approximate eigenvectors v_i from the Lanczos vectors q_i and the eigenvectors t_i of the tridiagonal matrix T according to

$$v_i = Q t_i \quad (14)$$

where Q is a matrix with q_i as its columns. In practice, we match the three biggest coefficients of each v_i for the various basis sets. Finally, since the number of satellites becomes very large when increasing the size of the basis set, we only extrapolate energies corresponding to satellites with significant spectral weight.

V. RESULTS

In this section we will assess the accuracy and feasibility of (3,1)-MCDE calculations for atoms and molecules. To judge the accuracy we will compare spectral functions obtained with the (3,1)-MCDE to those obtained within full configuration interaction (FCI) for a small basis set. Then, to illustrate the feasibility of (3,1)-MCDE calculations, we will calculate the MCDE spectral functions for large basis sets. Moreover, we will present a simple approach to obtain the MCDE spectral function in the complete basis-set limit.

A. (3,1)-MCDE vs. FCI and GW

For a given basis set, the FCI approach yields the exact ground and excited-state energies and wave functions. From these solutions the exact spectral function corresponding to that basis set can be obtained. Since FCI calculations scale exponentially with the number of basis functions, only calculations with few basis functions are numerically feasible. Nevertheless, for small basis sets, FCI provides a useful reference against which the accuracy of the (3,1)-MCDE spectral functions can be benchmarked.

In Fig. 1 we report the spectral functions of Neon and several small molecules obtained within the (3,1)-MCDE and compare them to the spectral functions obtained within FCI and *GW*. In the case of Neon we also compare to a spectral function from Ref. [45] that was obtained using the coupled-cluster approach to the single-particle Green's function with single and double excitations (GFCCSD) [32–34]. To facilitate the comparison, all spectra are aligned at the quasiparticle peak with the smallest binding energy.

For Neon we observe that the (3,1)-MCDE spectral function is in overall good agreement with the FCI spectrum and it is very similar to that obtained within GFCCSD. The (3,1)-MCDE spectrum provides a remarkable improvement over the *GW* spectral function which strongly overestimates the binding energy of the satellites. For example, the satellite with the lowest binding energy in the FCI spectrum is at -51.2 eV while the (3,1)-MCDE predicts it to be at -53.3 eV. Instead, *GW* predicts this satellite to be at -74.6 eV, an error of more than 23 eV.

The spectral functions for the small molecules show a similar trend, i.e., *GW* overestimates the binding energy of the satellites for all molecules except the smallest one, LiH, where it strongly underestimates them. Instead, the (3,1)-MCDE yields spectral functions that are overall closer to those obtained within FCI, although also the (3,1)-MCDE overestimate the satellite binding energies. In particular, for LiH and BeH₂ the MCDE spectra compare very well to the FCI spectra. Instead, the spectral functions obtained for HF, NH₃, and, especially, H₂O show larger differences with those obtained within FCI. Nevertheless, the (3,1)-MCDE spectral functions are overall in better agreement with the FCI spectra than those obtained with *GW*.

The satellite binding energies in the (3,1)-MCDE spectral functions are overestimated because not all the electron-hole and particle-particle interactions included in the three-particle self-energy Σ^{3p} are screened. Since screening effects are approximately proportional to the number of electrons, this omission has important consequences for large molecules. As we recently showed [44], all interactions can be properly screened by using the following three-particle self-energy

$$\begin{aligned} \Sigma_{ijl,mok}^{3p,scr} = & [(1-f_i)(1-f_j)f_l - f_if_j(1-f_l)] \\ & \times [\delta_{lk}(W_{ijom} - W_{ijmo}) + \delta_{mj}(W_{iklo} - v_{ikol}) \\ & + \delta_{io}(W_{jklm} - v_{jkml}) - \delta_{oj}(W_{iklm} - v_{ikml}) \\ & - \delta_{im}(W_{jklo} - v_{jkol})]. \end{aligned} \quad (15)$$

in which the matrix elements of the statically screened

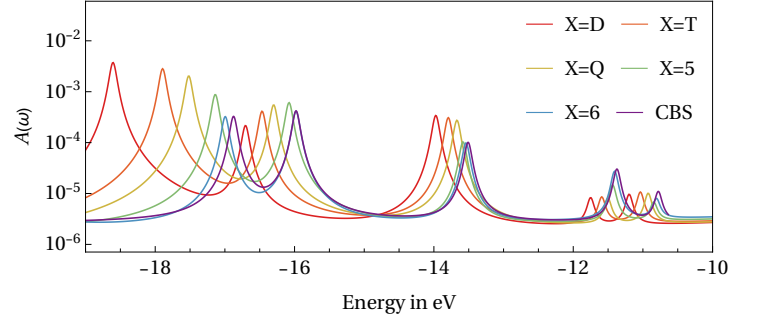


FIG. 2. The spectral function of the 5 neon 2s satellite peaks with lowest binding energy calculated using various aug-cc-pVXZ basis sets. We note that the spectral function is in logarithmic scale.

Coulomb potential $W(\mathbf{r}_1, \mathbf{r}_2)$ are given by

$$W_{ijlm} = \iint dx_1 dx_2 \phi_i^*(x_1) \phi_j^*(x_2) W(\mathbf{r}_1, \mathbf{r}_2) \phi_l(x_2) \phi_m(x_1). \quad (16)$$

By properly screening all electron-hole and particle-particle interactions this screened three-particle self-energy will reduce the binding energies of the satellites. This, however, is beyond the scope of this work. We note that the screened potential has also been employed to accurately describe other types of many-body excitations. [51–55].

So far, we have only used small basis sets because we wanted to compare to FCI spectra for which large basis sets are prohibitive. In the next section we will calculate the MCDE spectral functions using larger basis sets and demonstrate how one can obtain photoemission spectra in the complete basis-set limit.

B. Complete basis set limit

The binding energies and intensities of the satellites converge extremely slowly with the size of the basis set. As an example, in Fig. 2 we report the spectral function of Neon for

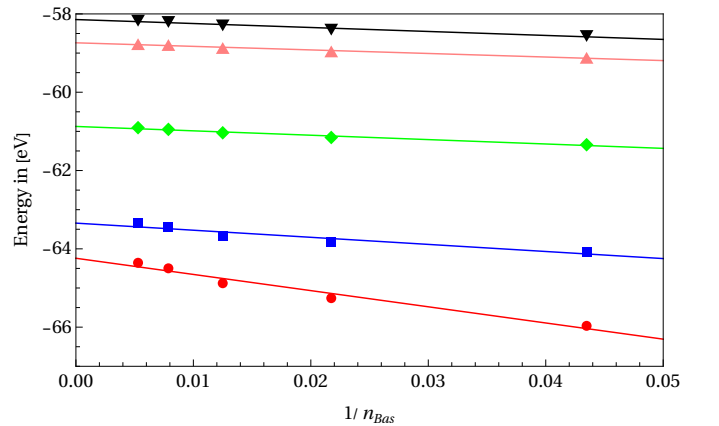


FIG. 3. Satellite energies of Neon 2s peak (Figure 4) for different basis sets, plotted against the inverse of the basis set size.

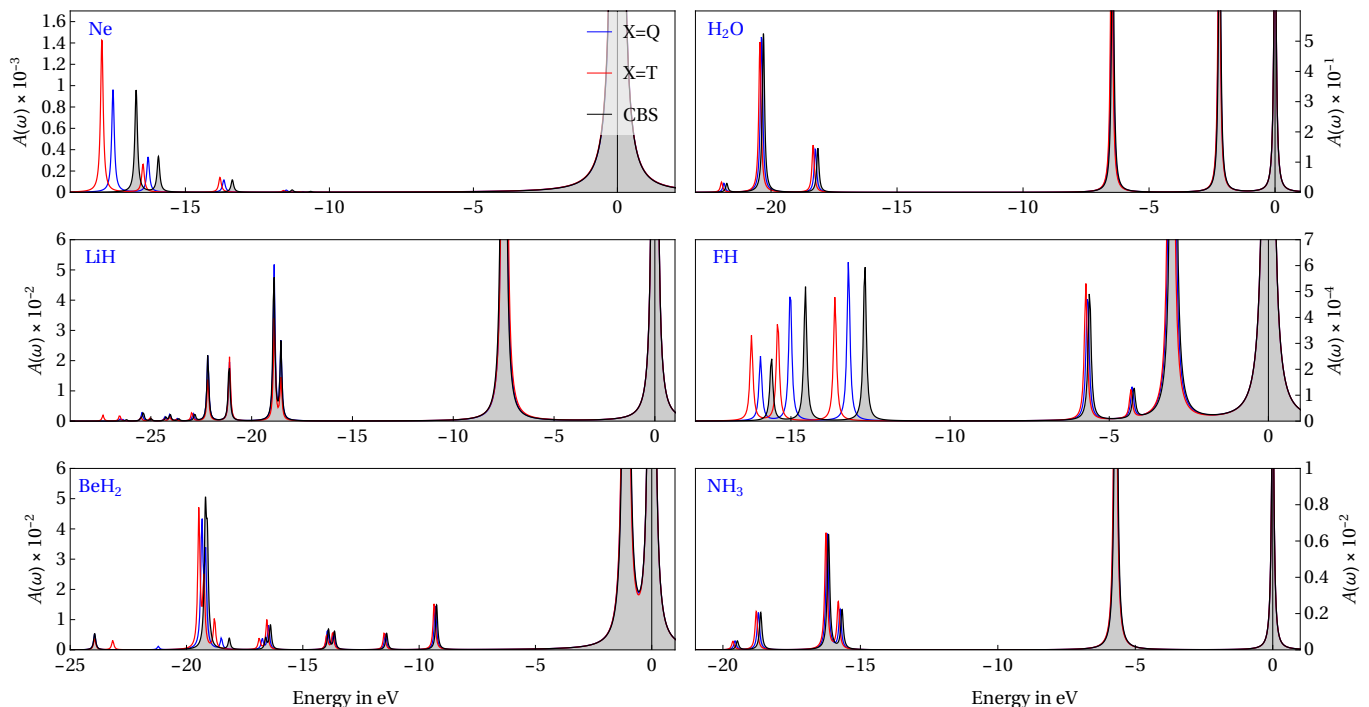


FIG. 4. The valence spectral functions of Neon and several molecules obtained by extrapolating to the complete basis-set limit using the aug-cc-pVQZ and aug-cc-pVTZ basis.

the aug-cc-pVXZ family of basis sets from double ζ ($X=D$) to sextuple ζ ($X=6$). We only show the part of the spectrum containing the five satellite energies with the smallest binding energies. We see that the closer a satellite is to its corresponding quasi-particle peak the faster is the convergence with basis-set size. This was to be expected since low-lying satellites involve higher virtual orbitals. However, even with the aug-cc-pV6Z basis set this part of the spectrum has not fully converged. Since there are many more lower-lying satellites that are not included in Fig. 2, these observations lead to the conclusion that extremely large basis sets are required to converge the full spectral function. However, calculations with such large basis sets are generally out of reach for most atoms and molecules.

Therefore, we propose a numerically much more efficient solution. It has been shown that quasiparticle energies behave linearly as a function of the inverse of the number of basis functions and that, therefore, the complete basis set (CBS) limit for these energies can be obtained by extrapolation [56]. Since both quasi-particle energies and satellite energies are electron removal and addition energies, a similar behaviour could exist for the satellite energies. To verify this idea, we plot the energies of the five Neon satellite energies of Fig. 2 as a function of the inverse of the number of basis functions. We report the results in Fig. 3 and we clearly observe that the satellite energies indeed lie on straight lines. Therefore, as for the quasiparticle energies, we can also extrapolate the satellite energies to the CBS limit. In Fig. 2 we reported the spectral function of Neon containing five satellites in the CBS limit. For the corresponding intensities of the satellite peaks,

TABLE I. Five Neon 2s satellite energies obtained from the extrapolation of the energies obtained using the aug-cc-pVXZ basis set for two consecutive values of X .

X	Neon 2s satellite energies (eV)				
D+T	-64.55	-63.59	-60.97	-58.79	-58.24
T+Q	-64.36	-63.45	-60.87	-58.75	-58.15
Q+5	-63.85	-63.05	-60.80	-58.66	-58.06
5+6	-64.07	-63.14	-60.82	-58.73	-58.11

the intensities obtained within the largest basis set has been used.

In Table I we report five Neon 2s satellite energies obtained from the extrapolation of two energies obtained with aug-cc-pVXZ basis sets using two consecutive values of X . We observe that the extrapolated values are all quite close. In practice, for the systems studied in this article, one can reach the CBS limit for the spectral functions with sufficient accuracy by extrapolating quasiparticle and satellite energies obtained within the aug-cc-pVTZ and aug-cc-pVQZ basis sets. In Fig. 4 we report the (3,1)-MCDE valence spectral functions thus obtained for Ne, LiH, BeH₂, H₂O, FH and NH₃. We observe that, for the given energy ranges, the Neon and FH spectral functions are not converged using the aug-cc-pVQZ basis and an extrapolation is crucial.

Finally, in Fig. 5 we report the (3,1)-MCDE core spectral functions for Ne, LiH, BeH₂, H₂O, FH and NH₃ obtained using the same extrapolation method. Again we observe that, for the given energy ranges, the spectral functions of some systems and, in particular Neon, are not converged using the

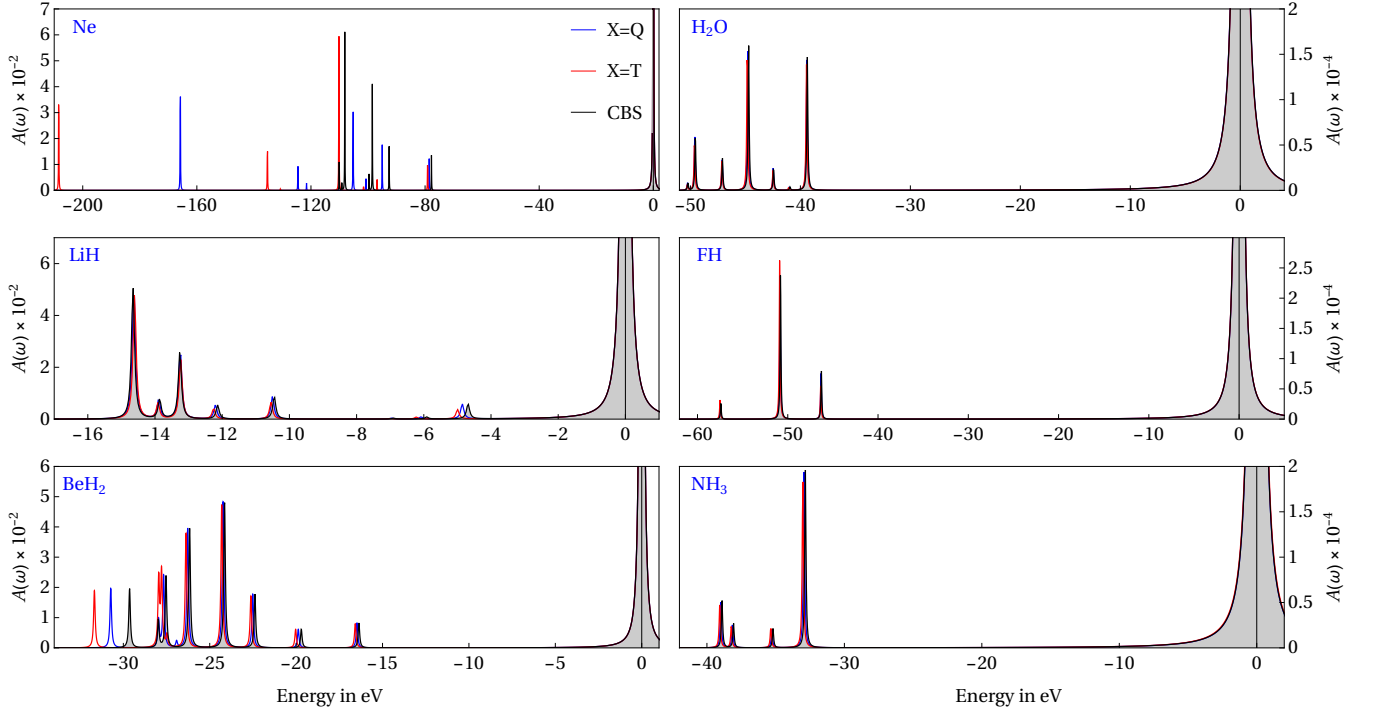


FIG. 5. The core spectral functions of Neon and several molecules obtained by extrapolating to the complete basis-set limit using the aug-cc-pVQZ and aug-cc-pVTZ basis.

aug-cc-pVQZ basis and an extrapolation is needed.

VI. CONCLUSIONS

We have implemented the (3,1)-MCDE for finite systems and used it to calculate the photoemission spectral functions of atoms and molecules. For small basis sets the (3,1)-MCDE spectra are in better agreement with FCI spectra than those obtained with *GW*. However, for some molecules significant differences remain for the binding energies of the satellites. This is due to a lack of screening of the electron-hole and particle-particle excitations in the (3,1)-MCDE. In future work we will investigate the performance of the screened (3,1)-MCDE, in which these excitations are properly screened, for the spectral functions of atoms and molecules.

We have also demonstrated that the binding energies of the satellites converge extremely slowly with the size of the basis set. We have presented a simple extrapolation technique to obtain spectral functions in the complete basis-set limit.

Finally, our implementation of the (3,1)-MCDE for atoms and molecules sets the stage for the implementation of other multichannel Dyson equations that describe other types of spectroscopies [40, 42].

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Appendix A: Spin-adapted (3,1)-MCDE equations

As mentioned in the main text, the three-particle channel of the effective Hamiltonian in Eq. 1 contains contributions from both spin doublets and spin quartets. However, only the doublet contributions couple to the one-particle channel since the latter only contains spin doublets. In order to separate the doublet and quartet states we can perform a basis transformation. For the spin up channel this transformation is given by [57]

$$A_i^D = A_{i\uparrow} = \varepsilon_i \quad (\text{A1})$$

$$A_{ijl}^{D1} = \frac{1}{\sqrt{2}}(A_{i\uparrow j\downarrow l\downarrow} - A_{i\downarrow j\uparrow l\downarrow}) \quad (\text{A2})$$

$$A_{ijl}^{D2} = \frac{1}{\sqrt{6}}(2A_{i\uparrow j\uparrow l\uparrow} + A_{i\uparrow j\downarrow l\downarrow} + A_{i\downarrow j\uparrow l\downarrow}) \quad (\text{A3})$$

$$A_{ijl}^Q = \frac{1}{\sqrt{3}}(-A_{i\uparrow j\uparrow l\uparrow} + A_{i\uparrow j\downarrow l\downarrow} + A_{i\downarrow j\uparrow l\downarrow}) \quad (\text{A4})$$

$$(\text{A5})$$

Within this basis, the spin-up channel of the effective Hamiltonian for the doublet states $H_3^{D,\uparrow}$ becomes

$$H_3^{D,\uparrow} = \begin{pmatrix} \varepsilon^{1p} & C^{D1} & C^{D2} \\ [C^{D1}]^\dagger & \varepsilon^{3p} + H^{D1} & C^{D1,D2} \\ [C^{D2}]^\dagger & [C^{D1,D2}]^\dagger & \varepsilon^{3p} + H^{D2} \end{pmatrix} \quad (\text{A6})$$

in which

$$\epsilon_{i,m}^{1p} = \delta_{im} \epsilon_i \quad (\text{A7})$$

$$\epsilon_{i \geq j; m \geq ok}^{3p} = \delta_{im} \delta_{jo} \delta_{lk} (\epsilon_i + \epsilon_j - \epsilon_l) \quad (\text{A8})$$

$$C_{i,m \geq ok}^{D1} = \left[\sqrt{\frac{1}{2}} \right]^{\delta_{mo}} \sqrt{\frac{1}{2}} (f_m - f_k) (f_o - f_k) [\mathcal{V}_{ikom} + \mathcal{V}_{ikmo}] \quad (\text{A9})$$

$$C_{i,m \geq ok}^{D2} = \sqrt{\frac{3}{2}} (f_m - f_k) (f_o - f_k) [\mathcal{V}_{ikom} - \mathcal{V}_{ikmo}] \quad (\text{A10})$$

$$[C^{D1}]_{i \geq j; l; m}^\dagger = \left[\sqrt{\frac{1}{2}} \right]^{\delta_{ij}} \sqrt{\frac{1}{2}} (f_i - f_l) (f_j - f_l) [\mathcal{V}_{ijlm} + \mathcal{V}_{ijml}] \quad (\text{A11})$$

$$[C^{D2}]_{i \geq j; l; m}^\dagger = \sqrt{\frac{3}{2}} (f_i - f_l) (f_j - f_l) [\mathcal{V}_{ijlm} - \mathcal{V}_{ijml}] \quad (\text{A12})$$

$$H_{i \geq j; l; m \geq ok}^{D1} = \left[\sqrt{\frac{1}{2}} \right]^{\delta_{mo}} \left[\sqrt{\frac{1}{2}} \right]^{\delta_{ij}} [(1-f_i)(1-f_j)f_l - f_i f_j (1-f_l)] \left(\delta_{lk} (\mathcal{V}_{ijom} + \mathcal{V}_{ijmo}) - \delta_{mj} (\mathcal{V}_{iklo} - \frac{1}{2} \mathcal{V}_{ikol}) \right. \\ \left. - \delta_{io} (\mathcal{V}_{jklm} - \frac{1}{2} \mathcal{V}_{jkml}) - \delta_{oj} (\mathcal{V}_{iklm} - \frac{1}{2} \mathcal{V}_{ikml}) - \delta_{im} (\mathcal{V}_{jklo} - \frac{1}{2} \mathcal{V}_{jkol}) \right) \quad (\text{A13})$$

$$H_{i \geq j; l; m \geq ok}^{D2} = [(1-f_i)(1-f_j)f_l - f_i f_j (1-f_l)] \left(\delta_{lk} (\mathcal{V}_{ijom} - \mathcal{V}_{ijmo}) + \delta_{mj} (\mathcal{V}_{iklo} \right. \\ \left. - \frac{3}{2} \mathcal{V}_{ikol}) + \delta_{io} (\mathcal{V}_{jklm} - \frac{3}{2} \mathcal{V}_{jkml}) - \delta_{oj} (\mathcal{V}_{iklm} - \frac{3}{2} \mathcal{V}_{ikml}) - \delta_{im} (\mathcal{V}_{jklo} - \frac{3}{2} \mathcal{V}_{jkol}) \right) \quad (\text{A14})$$

$$C_{i \geq j; l; m \geq ok}^{D1, D2} = \left[\sqrt{\frac{1}{2}} \right]^{\delta_{ij}} \frac{\sqrt{3}}{2} [(1-f_i)(1-f_j)f_l - f_i f_j (1-f_l)] [\delta_{mj} \mathcal{V}_{ikol} - \delta_{io} \mathcal{V}_{jkml} - \delta_{oj} \mathcal{V}_{ikml} + \delta_{im} \mathcal{V}_{jkol}] \quad (\text{A15})$$

$$[C^{D1, D2}]_{i \geq j; l; m \geq ok}^\dagger = \left[\sqrt{\frac{1}{2}} \right]^{\delta_{mo}} \frac{\sqrt{3}}{2} [(1-f_i)(1-f_j)f_l - f_i f_j (1-f_l)] [-\delta_{mj} \mathcal{V}_{ikol} + \delta_{io} \mathcal{V}_{jkml} - \delta_{oj} \mathcal{V}_{ikml} + \delta_{im} \mathcal{V}_{jkol}]. \quad (\text{A16})$$

Here all indices refer to spatial orbitals and where

$$\mathcal{V}_{ijlm} = \iint dr_1 dr_2 \phi_i^*(\vec{r}_1) \phi_j^*(\vec{r}_2) v(\vec{r}_1, \vec{r}_2) \phi_l(\vec{r}_2) \phi_m(\vec{r}_1), \quad (\text{A17})$$

in which $\phi_i(\vec{r})$ is a Hartree-Fock spatial orbital. A similar transformation can be done for the spin-down channel.

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